

MIDTERM EXAM

ECO403/503–INTRODUCTION TO CONTRACT THEORY & AUCTIONS

SEPTEMBER 30, 2025

Name

ID

Instructions: The exam has a total of 100 points. All questions have the same weight. The duration of the exam is 1 hour and 15 minutes (75 minutes).

This booklet contains both the questions and the space for your answers. You must write all of your answers directly in the spaces provided. **You may not use extra sheets of paper for your answers.** Scratch paper can be provided upon request, but anything written there will not be graded.

Do not remove the staple from this booklet.

Please write legibly and concisely. Clear and organized presentation is essential. When explaining your argument, do not just present the math. Write in complete sentences and explain what you are doing. Cross out anything you do not want me to consider.

Fill this cover sheet with your name and student ID before beginning. At the end of the exam, return this entire booklet with both questions and answers.

Problem 1	Problem 2	Problem 3	Problem 4	Total

Problem 1: Pokémon Duel with Spillovers

Our two TAs, Sultan and Bagdat, duel for a rare shiny Pokémon card worth $W > 0$. Each chooses non-negative effort x_i (deck building, trading, scrimming). The probability that player i wins the duel is given by

$$p_i(x_i, x_j) = \begin{cases} \frac{x_i}{x_i + x_j} & \text{if } x_i + x_j \neq 0, \\ 1/2 & \text{if } x_i + x_j = 0. \end{cases}$$

Correspondingly, the probability that i loses the duel is given by $1 - p_i(x_i, x_j)$.

However, in the world of Pokémon dueling, battles have spillovers: even if you lose, you might still gain something from your rival's effort, e.g., learning new strategies, increasing your reputation in the circuit, picking up booster packs, etc.

Formally, the payoff of player i is:

$$u_i(x_i, x_j) = \begin{cases} W - x_i + \beta_1 x_j & \text{if } i \text{ wins,} \\ -x_i + \beta_2 x_j & \text{if } i \text{ loses,} \end{cases}$$

where $\beta_1 \geq 0$ and $\beta_2 \geq 0$ are parameters which capture the extent of Pokémon-card battle spillovers.

- 1.1. Write down the expected payoff of player i .
- 1.2. Compute the player i 's best response function. State a sufficient condition on the parameters under which the maximization problem of player i is strictly concave (and thus has a unique global maximizer).
- 1.3. Derive the symmetric Nash equilibrium.
- 1.4. How do the equilibrium efforts depend on the parameters β_1 and β_2 ? *Hint: think about whether equilibrium effort rises when the benefit from winning spillovers increases (β_1), and whether it falls when the loser's spillover benefit (β_2) increases.*

Begin your solution for Problem 1 here.

Continue solution for Problem 1 here.

Problem 2: The Guessing Game Experiment

Fifty students are recruited for an important experiment designed by Prof. Alejandro, who somehow secured a grant for this and managed to bypass human-ethical experimental restrictions in the process.

Each student writes one integer between 0 and 100 on a slip of paper, alongside their name. Prof. Alejandro collects all the slips, and calculates the average x of all the numbers written by the students. The winner of the game is the student (or students) who wrote down the number that is closest to $\frac{2}{3}x$. The winners equally divide the prize of \$1000 USD between them.

- 2.1. Describe this game as a static game in normal-form: specify the players, their action sets, and their payoff functions.
- 2.2. Find all Nash equilibria of the game.
- 2.3. If you were one of the students, what number would you write on your slip, and why?

Begin your solution for Problem 2 here.

Continue solution for Problem 2 here.

Problem 3: The Coordinated TA-Attack Problem

Two divisions of the CAT-TA Army, with Akmaral leading the left wing and Sezim commanding the right wing, are secretly plotting a bold maneuver: to jump on their Head TA, Abuzer, in the NU atrium. If both strike at the same time, the rebellion succeeds, and Abuzer will finally let them do the specific tutorials they want. But if only one of them jumps while the other hesitates, the lone attacker will be captured and sentenced to Abuzer's ultimate punishment: being sent to hang out with the business administration bros.

Akmaral suggests a coordinated jump tomorrow at **10:00 AM sharp**, while Abuzer is going to one of his lectures. But neither she nor Sezim will dare to attack Abuzer unless they are absolutely sure the other is also attacking. After all, nobody wants to endure the torture of hanging out with the business bros.

Since Abuzer controls all the Telegram channels, the only way Akmaral and Sezim can communicate is through their friend Nozima, who acts as a courier for the messages. Nozima takes 10 minutes to get from one TA to the other. Abuzer usually patrols the courier routes, which also exposes Nozima to capture and the same torturous punishment of being sent to hang out with the business bros. Luckily, tonight Abuzer is busy preparing his PhD applications, so he is not patrolling the courier route.

Question: How long will it take before Akmaral and Sezim can actually coordinate the jump on their Head TA, Abuzer, inside NU? Justify your answer.

Begin your solution for Problem 3 here.

Problem 4: Aumann's Agreement Theorem

Consider the following version of Aumann's Agreement Theorem.

Theorem. Let $(N, \Omega, (\mathcal{P}_i)_{i \in N}, \mu)$ be a finite belief space with common prior μ such that $\mu(\omega) > 0$ for all $\omega \in \Omega$. Suppose that $N = \{1, 2\}$. Let $A \subseteq \Omega$ be an event and let $\omega \in \Omega$ be a state of the world. If the event "Player 1 ascribes probability q_1 to A " is common knowledge, and the event "Player 2 ascribes probability q_2 to A " is also common knowledge in ω , then $q_1 = q_2$.

Below, I outline an argument that attempts to prove this result in a different way than the one we saw in class. The argument is summarized in 6 steps, and certain details are missing. In particular, each step contains an underlined statement that needs verification. Either prove these statements, or give a precise reason(s) why you would expect them to be false.

- (i) Let C be the connected component of ω in the graph corresponding to the belief space above.
It follows that C is common knowledge at ω .
- (ii) $C = \bigcup_j P_1^j$, where $P_1^j \in \mathcal{P}_1$ for each j .
- (iii) $\mu(P_1^j) > 0$ for every j , and therefore $\mu(C) > 0$.
- (iv) The fact that Player 1 ascribes probability q_1 to the event A is common knowledge in ω .
Then for each j , $\mu(A \mid P_1^j) = \frac{\mu(A \cap P_1^j)}{\mu(P_1^j)} = q_1$.
- (v) It follows that $\mu(A \cap C) = q_1 \mu(C)$.
- (vi) We similarly derive that $\mu(A \cap C) = q_2 \mu(C)$. Since $\mu(C) > 0$, the previous steps imply $q_1 = q_2$.

Begin your solution for Problem 4 here.

Continue solution for Problem 4 here.